

Basal mixing model: forward dynamics and sampling strategies

BasalMixing repository

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Abstract

This document describes the 1-D basal mixing model used in the BasalMixing repository to interpret deep-ice ^{81}Kr and $\delta^{40}\text{Ar}_{\text{atm}}$ measurements at GISP2, and the Markov-chain Monte Carlo (MCMC) strategy used to jointly infer the mixing parameters and the time t_0 of glacial inception at the GISP2 site. The recommended strategy (`:sampled`) treats t_0 as a physical parameter of interest with a uniform prior on $[-3000, -700]$ kyr, where the bounds come from the geological window of first Greenland Ice Sheet (GrIS) glaciation (~ 2700 kyr ago) and the lower bound on basal ice age implied by ^{81}Kr measurements assuming no mixing (~ 700 kyr).¹ Two earlier strategies (`:profile` and `:marginal`) are retained for comparison and historical context; they are not the recommended treatment.

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1 The basal mixing model

1.1 Physical setting

GISP2 retrieves ice from the bottom of the Greenland Ice Sheet. The deepest ~ 13 m of the core sits above the bedrock and, unlike the rest of the column, is thought to have been mechanically and diffusively mixed with material below. Specifically: above $z = z_{\text{lim}} \approx 3040$ m the ice is

¹Sign convention: t_0 is reported in signed kyr (negative = past); $t_0 = 0$ is the present day.

treated as undisturbed (“clean” ice); below z_{lim} , ice is increasingly contaminated by debris-rich (“dirty”) material that acts as a sink/source for the tracers we measure. The model treats this lower layer as a 1-D advection–diffusion column extending to a fixed bedrock at $z_{\text{bed}} = 3053.44 \text{ m}$.

1.2 State variables and grid

The model carries two state variables at every cell centre z_j in a finite-volume discretization:

- $c_{81}(z, t)$: relative ^{81}Kr concentration, dimensionless. Initialized to 1 at $t = 0$ (modern abundance) at every depth.
- $c_{40}(z, t)$: $\delta^{40}\text{Ar}_{\text{atm}}$ concentration, in cc m^{-3} . Initialized to the modern equilibrium value $A_{40}^0 = (100 \text{ cm})^3 \cdot a_{\text{tac}} \cdot f_{\text{atm}} \approx 8000 \text{ cc/m}^3$, where $a_{\text{tac}} = 0.08$ is the typical air content and $f_{\text{atm}} = 0.00934$ is the present-day atmospheric $\delta^{40}\text{Ar}_{\text{atm}}$ volume fraction.

The model integrates the forward dynamics from glacial inception at t_0 up to the present day $t_{\text{end}} = 0$. The internal time variable used by the integrator is the positive elapsed time $\tau = t - t_0 \in [0, |t_0|]$, advanced with a forward Euler step $d\tau = 0.2 \text{ kyr}$ (with an automatic retry at $d\tau/2$ if the integrator produces non-physical, negative concentrations). The dynamics are autonomous in τ , so the only information the integrator needs about absolute time is the total duration $|t_0|$.

1.3 Mixing rate profile

The depth-dependent mixing rate is parameterised by two values m_{clean} and $m_{\text{dirty}} = f_{\text{dirty}} m_{\text{clean}}$ joined by a smooth transition of thickness δ centred at z_{lim} . Using w_1 and w_2 for the two tanh-shaped step functions,

$$w_1(z) = \frac{1}{2} \left(1 + \tanh\left(\frac{\beta}{\delta}(z - z_{\text{lim}})\right) \right), \quad (1)$$

$$w_2(z) = \frac{1}{2} \left(1 + \tanh\left(\frac{\beta}{\delta}(z - z_{\text{lim}} - \delta)\right) \right), \quad (2)$$

$$\Phi(z) = m_{\text{dirty}} w_1(z) w_2(z) + m_{\text{clean}} w_1(z) (1 - w_2(z)), \quad (3)$$

with sharpness $\beta = 50$ (essentially a step on the discretisation scale). So $\Phi = 0$ in clean ice, $\approx m_{\text{clean}}$ in the transition zone of width δ , and $\approx m_{\text{dirty}}$ in the dirty zone below.

The diffusivity is $D(z) = \Phi(z) L_{\text{ref}}$ with $L_{\text{ref}} = 1 \text{ m}$ treated as a reference length (it just rescales m_{clean} ; we keep $L_{\text{ref}} = 1$ and let m_{clean} carry the magnitude).

1.4 Governing equations

The continuum dynamics of $c \in \{c_{81}, c_{40}\}$ are

$$\frac{\partial c}{\partial \tau} = \frac{\partial}{\partial z} \left(D(z) \frac{\partial c}{\partial z} \right) + S_c(z, \tau), \quad (4)$$

discretised cell-centre to cell-centre, with τ the positive elapsed time since inception. The source/sink term S_c differs by tracer:

- For ^{81}Kr , $S_{81}(z, \tau) = -\lambda c_{81}(z, \tau)$ with $\lambda = \ln 2/t_{1/2}$ and $t_{1/2} = 229 \text{ kyr}$, except in the clean-ice cells for elapsed times $\tau > t_{\text{old}}$, where the decay term is switched off. Physically, t_{old} is the residence time of the clean ice column above the mixing zone: ice older than t_{old} is assumed to have been flushed out by ice flow, so its ^{81}Kr clock is capped at t_{old} .

- For $\delta^{40}\text{Ar}_{\text{atm}}$, $S_{40} = 0$ in the interior; at the bedrock cell only, a flux F_{40} is added:
 $S_{40}(z_{\text{bed}}, \tau) = F_{40}/\Delta z_{\text{bed}}$.

The observable quantities derived from the state are

$$\tau_{81}(z, \tau) = -t_{1/2} \log_2(c_{81}(z, \tau)), \quad \text{Kr-81 closed-system age, kyr}, \quad (5)$$

$$\delta^{40}\text{Ar}_{\text{atm}}(z, \tau) = 1000 \left(\frac{c_{40}(z, \tau)}{A_{40}^0} - 1 \right) - 0.066 t_{\text{old}}, \quad \delta^{40}\text{Ar, } \%, \quad (6)$$

where the $-0.066 t_{\text{old}}$ term corrects for atmospheric $\delta^{40}\text{Ar}_{\text{atm}}$ drift over the residence time of the clean column (Bender et al. 2010).

1.5 Parameters

The six sampled parameters and the two fixed observation-noise scales are:

Symbol	Meaning	Units	Prior	Code name
δ	transition-zone thickness	m	Uniform(0.3, 2.0)	delta
m_{clean}	mixing rate in transition zone	m/kyr	Normal(0.03, 0.002) trunc. ≥ 0	m_clean
f_{dirty}	dirty-to-clean mixing-rate ratio	—	Uniform(4.0, 6.5)	f_dirty
t_{old}	clean-column residence time	kyr	Normal(250, 25) trunc. ≥ 0	t_old
F_{40}	bedrock $\delta^{40}\text{Ar}_{\text{atm}}$ source flux	cc/m ² /kyr	Uniform(0.004, 0.007)	F_ar40
t_0	glacial inception time	kyr (signed)	Uniform(−3000, −700)	t_0
σ_{81}	^{81}Kr age obs. noise (<i>fixed</i>)	kyr	30	σ_{k81}
σ_{40}	$\delta^{40}\text{Ar}$ obs. noise (<i>fixed</i>)	‰	0.03	σ_{dar40}

The choice to fix σ_{81} and σ_{40} rather than place priors on them was made to improve sampling efficiency. Both values are close to the analytical errors of the measurements; relaxing them to half-Cauchy or exponential priors is a minor change to the model if needed.

1.6 Observations

The model is compared against two data sets, both at observation depths z_i corresponding to specific GISP2 ice samples:

- Three ^{81}Kr closed-system ages, τ_{81}^{obs} , at depths 3045–3050 m (typically reported as ages of 577, 649, and 714 kyr).
- Ten $\delta^{40}\text{Ar}_{\text{atm}}$ measurements from Bender et al. (2010), spanning depths 3036.5 to 3052.4 m, with values rising from ~ 0 ‰ in the clean zone to ~ 0.5 ‰ near bedrock.

For depths between cell centres the model predictions are linearly interpolated. Both data sets are assumed to be independent Gaussian observations of the corresponding model quantities at the depths in question.

2 The likelihood and the inception-time problem

Given parameters $\theta = (\delta, m_{\text{clean}}, f_{\text{dirty}}, t_{\text{old}}, F_{40})$ and an inception time t_0 , the model integrates over the elapsed duration $|t_0|$ to produce present-day predictions $\hat{\tau}_{81}(z_i; \theta, t_0)$ and $\widehat{\delta^{40}\text{Ar}}(z_j; \theta, t_0)$ at the observation depths z_i, z_j . Conditional on θ and t_0 , the log-likelihood is the usual Gaussian:

$$\log L(\theta, t_0 \mid \mathbf{y}) = -\frac{1}{2\sigma_{81}^2} \sum_i (\hat{\tau}_{81}(z_i; \theta, t_0) - y_i^{81})^2 - \frac{1}{2\sigma_{40}^2} \sum_j (\widehat{\delta^{40}\text{Ar}}(z_j; \theta, t_0) - y_j^{40})^2 + C, \quad (7)$$

where C collects θ - and t_0 -independent normalising constants.

The inference problem is to recover the joint posterior on (θ, t_0) given the GISP2 data $\mathbf{y}$.

Physical bounds on t_0 . Two independent lines of evidence bracket t_0 :

- *Upper bound ($t_0 \geq -3000$ kyr):* Greenland was essentially ice-free before ~ 2700 kyr ago. Inception of the GrIS at the summit cannot have occurred earlier than the onset of large-scale Northern-Hemisphere glaciation, which the geological record places no earlier than the late Pliocene.
- *Lower bound ($t_0 \leq -700$ kyr):* the deepest ^{81}Kr sample carries a closed-system age of ~ 714 kyr. Treated as a no-mixing model (i.e. as the in-situ ^{81}Kr clock without any rejuvenation from below), that age is a lower bound on how long the basal ice has been sitting at the summit. Any mixing only *adds* atmospheric-fresh ^{81}Kr from above, so the true ice age must be at least that old.

These two bounds motivate the prior $t_0 \sim \text{Uniform}(-3000, -700)$ kyr used in the recommended (`:sampled`) inference.

Data-constraint structure. The data weakly constrain t_0 from *within* its prior window. The basal mixing system reaches an effective equilibrium on timescales $t_{\text{eq}}(\theta) \sim 1000$ kyr or less, after which the present-day predictions are nearly independent of how much longer the system has been running. Two inception times $|t_0^{(1)}|, |t_0^{(2)}| > t_{\text{eq}}(\theta)$ produce essentially identical predictions and identical likelihoods. This saturation is the central feature of the inference and explains why the posterior on t_0 is wide: the data alone cannot tell us that the GrIS inceptioned at -2300 vs. -1500 kyr if both durations comfortably exceed the equilibration time. What *can* be ruled out is inception times shorter than t_{eq} , where the system has not yet relaxed.

3 Sampling strategies

The repository implements three strategies, selectable by the `model_kind` variable at the top of `run_basalmixing_ensemble.jl`. All three use the same affine-invariant ensemble sampler (Emcee; Goodman & Weare 2010) with 32 walkers, so their wall-clock costs are directly comparable per integration of the forward model.

The recommended strategy is `:sampled`. The other two (`:profile` and `:marginal`) predate the present framing and are retained for diagnostic comparison; their limitations are discussed below.

3.1 `:sampled` — joint (θ, t_0) posterior (*recommended*)

What it does. Treat t_0 as one of the model’s physical parameters, with the prior $t_0 \sim \text{Uniform}(-3000, -700)$ kyr motivated in Section 2. For each (θ, t_0) proposal, run the forward model once over elapsed duration $|t_0|$ and evaluate (7) at the present-day endpoint.

Assumptions.

- t_0 is a meaningful physical quantity — the time of glacial inception at the GISP2 site — about which we have prior information from independent geological and isotopic constraints.
- The comparison time is the present day; the integration finishes exactly there.

What you get out. A proper joint Bayesian posterior on (θ, t_0) . The marginal on θ accounts for uncertainty in t_0 by averaging; the marginal on t_0 is the full inference about inception time given the data *and* the physical bounds.

Pros and cons. This is the appropriate Bayesian treatment given that we have meaningful prior information about t_0 and care about its posterior. The 6-dimensional chain mixes well under `Emcee` (acceptance ≈ 0.24 in test runs at 32 walkers, 500 iterations), with wall-clock per sample governed by the mean of $|t_0|$ across the prior.

The posterior on t_0 is honestly wide because the data have weak post-equilibration power, but the width is now physically interpretable: it represents how much the data narrow the prior window, not an artefact of an arbitrary upper integration bound (see the discussion of `:sampled-with-elapsed-time-prior` below for the previous, deprecated framing).

3.2 `:profile` — profile likelihood (*diagnostic only*)

What it does. For every parameter proposal θ , integrate the forward model from elapsed time $\tau = 0$ to a fixed upper bound $\tau_1 = 3000$ kyr on a 1-kyr grid. At each grid point compute the log-likelihood as if that τ were the duration $|t_0|$. Select the elapsed time that maximises the likelihood:

$$\tau^*(\theta) = \arg \max_{\tau} \log L(\theta, \tau \mid \mathbf{y}), \quad (8)$$

and use $\log L^*(\theta) = \log L(\theta, \tau^* \mid \mathbf{y})$ as the contribution to the MCMC log-posterior. The chain reports $t_0 := -\tau^*(\theta)$ as a derived (non-sampled) quantity for naming consistency with `:sampled`.

Assumptions.

- No prior on t_0 .
- The “correct” t_0 for each θ is treated as a deterministic function of θ , chosen to make the data fit best.

What you get out. A posterior on θ from a non-standard (non-Bayesian) likelihood: the *profile likelihood* $L^*(\theta)$. The reported t_0 per chain sample is a point optimum, not a sample from a posterior.

Pros and cons. Computationally efficient (the inner integration can early-terminate once predicted ages exceed the range of interest). *However:* the posterior on θ is biased — it over-credits each θ for the best t_0 available rather than averaging the data’s support over plausible t_0 . The reported t_0 “posterior” is a histogram of point optima with no probability-distribution interpretation. The prior information on t_0 ($-3000 \leq t_0 \leq -700$ kyr) is not used.

3.3 `:marginal` — elapsed time as a nuisance parameter (*deprecated*)

Historical context. The `:marginal` strategy was developed under an earlier framing in which the integration duration was treated as a nuisance parameter with no independent physical interpretation. In that framing, the goal was to infer θ alone, marginalising the duration out of the likelihood:

$$\log p(\mathbf{y} \mid \theta) = \log \int_0^{\tau_1} L(\theta, \tau \mid \mathbf{y}) p(\tau) d\tau, \quad (9)$$

with a uniform prior $p(\tau) = 1/\tau_1$ on $[0, \tau_1]$. Numerically the integral is evaluated as a trapezoidal sum on the 1-kyr prediction grid:

$$\log p(\mathbf{y} \mid \theta) \approx \log \left(\sum_k \exp(\log L(\theta, \tau_k)) \Delta\tau \right), \quad (10)$$

computed via the standard `logsumexp` trick to avoid underflow. An online equilibration check (compare predicted observation vectors $W = 100$ slices apart; declare equilibrium when both ^{81}Kr and $\delta^{40}\text{Ar}_{\text{atm}}$ predictions have moved less than $\sim 1\%$ of the corresponding observation noise) allows the integrator to skip the post-equilibration tail:

$$\log p(\mathbf{y} \mid \theta) \approx \log \left(\sum_{k \leq k_{\text{eq}}} \exp(\log L(\theta, \tau_k)) \Delta\tau + (\tau_1 - \tau_{\text{eq}}) \exp(\log L_{\text{eq}}) \right). \quad (11)$$

Why it is no longer recommended. The marginal-likelihood framing required t_0 to be a nuisance parameter with no independent prior information — a stance that we no longer hold. With the physical bounds derived in Section 2, t_0 carries a meaningful prior of width 2300 kyr, and its posterior is a quantity we care about (it tells us when the GrIS inceptioned at the summit). Two specific failure modes of `:marginal` under the present framing:

- *The prior on t_0 is discarded.* Equation (9) integrates against a uniform prior on the elapsed-time variable $[0, \tau_1 = 3000]$ kyr, which is not the same as the physical prior $t_0 \sim \text{Uniform}(-3000, -700)$. Inception times shorter than 700 kyr (excluded by the ^{81}Kr no-mixing argument) are still given weight in the integrand.
- *The posterior on t_0 is no longer in the chain.* A post-hoc reconstruction $p(t_0 \mid \mathbf{y}) \propto N^{-1} \sum_s L(\theta^{(s)}, t_0) / Z(\theta^{(s)})$ averages per-slice likelihood curves over the marginalised- θ chain. It is a valid summary of what the data say about t_0 under the uniform-on-elapsed-time prior, but it does not correctly incorporate the physical prior either, and it requires re-running the forward model at every chain draw to recover.

When to still use it. For sensitivity checks: comparing the `:sampled` posterior on θ against the `:marginal` posterior under a deliberately permissive prior tells you how much the physical prior is doing. They should agree broadly on the well-identified parameters (m_{clean} , t_{old}); a large disagreement on f_{dirty} , δ , or F_{40} would flag a posterior strongly shaped by the t_0 prior, which is worth knowing.

3.4 Comparison summary

	t_0 treatment	prior on t_0	posterior on t_0	status
<code>:sampled</code>	sampled parameter	$\text{Uniform}(-3000, -700)$	in chain	recommended
<code>:profile</code>	point optimum	none	histogram of optima	diagnostic
<code>:marginal</code>	integrated out	$\text{Uniform}(0, 3000)$ on τ	post-hoc reconstruction	deprecated

4 Practical considerations

4.1 Sampler and threading

All three model kinds use Emcee with 32 walkers and 500 iterations ($32 \times 500 = 16\,000$ likelihood evaluations). Emcee is appropriate here because the posterior is moderately multimodal and the forward model is not differentiable in a way that supports gradient-based samplers (it has a `try/catch` on `DomainError` and mutates buffers, breaking `ForwardDiff`).

Threading: each thread owns its own `BasalMixingModel` object, allocated to length `Threads.maxthreadid()` to accommodate Julia’s interactive threads in newer versions.

4.2 Numerical robustness

The forward Euler integrator at $\Delta\tau = 0.2\text{kyr}$ is marginally stable; for some proposals the explicit step overshoots and produces non-positive concentrations, raising `DomainError` in `concentration_to_age`. The forward function catches these and reports failure; the `@model` retries once at $\Delta\tau/2$ before giving up and attaching a vanishingly small likelihood.

4.3 Outputs and reproducibility

Each ensemble run writes a snapshot to `results/emcee-chain-<model_kind>.jld2`, containing the chain, the data `DataFrames`, the depth grid, the priors, the sampler and the model kind. The plotting helpers (`plot_basalmixing_ensemble.jl`) load this snapshot and re-derive figures, switching the MAP best-fit re-run on the recorded `model_kind`. For `:sampled` chains the best-fit re-run integrates exactly from t_0 to t_{end} (no extrapolation past the inference endpoint); for `:profile` and `:marginal` it runs the full $0 \rightarrow \tau_1$ dense grid and annotates the argmin slice.

References

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